

Legal and Security Challenges in Cyber-Physical Systems: A Framework for Secure and Compliant IoT Ecosystems

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ABSTRACT

Cyber-Physical Systems (CPS) have become the technological foundation of modern critical infrastructures, including smart grids, industrial automation, healthcare systems, intelligent transportation systems, and autonomous vehicles. By integrating physical processes with computational intelligence, communication networks, and sensing technologies, CPS enable real-time monitoring, automation, and intelligent decision-making. However, the rapid convergence of Artificial Intelligence (AI), Internet of Things (IoT), cloud computing, and operational technologies has significantly expanded the cyberattack surface of CPS ecosystems. As a result, cybersecurity incidents can lead not only to data breaches but also to physical damage, operational disruption, legal liability, regulatory penalties, and public safety risks. Existing research predominantly focuses on technical security mechanisms while providing limited attention to legal compliance, governance, privacy protection, and AI accountability. This paper addresses this gap by proposing the Secure and Compliant Cyber-Physical Systems Framework (SCCF), a multidisciplinary governance architecture integrating cybersecurity, data governance, AI governance, and legal-regulatory compliance. A



comparative analysis of the Information Technology Act, 2000, Digital Personal Data Protection Act, 2023, GDPR, NIST CSF 2.0, ISO/IEC 27001, and IEC 62443 is conducted to identify governance requirements relevant to CPS environments. The findings reveal that existing frameworks remain fragmented and fail to provide comprehensive guidance for integrating security and compliance. The proposed SCCF introduces a five-layer governance model consisting of Physical Security, Cybersecurity, Data Governance, AI Governance and Ethics, and Legal-Regulatory Compliance. The framework operationalizes Security-by-Design, Privacy-by-Design, Compliance-by-Design, and Governance-by-Design principles, enabling organizations to develop secure, trustworthy, and legally compliant CPS ecosystems.

1 Introduction

Cyber-Physical Systems (CPS) are part of Industry 4.0, which has accelerated their deployment within manufacturing, healthcare, transportation, energy and smart cities. CPS help enable intelligent operations and autonomous decision-making by integrating sensing technology, embedded computers, communication networks and physical processes.

Artificial Intelligence (AI) and Internet of Things (IoT) technologies have enhanced the capabilities of CPS through predictive maintenance, anomaly detection, resource optimisation, and autonomous control.

CPS environments face significant cybersecurity challenges that differ from traditional information systems; attacks against CPS directly impact physical infrastructure, resulting in disruption to operations, damage to equipment, environmental hazards and threats to the safety of humans. This is exemplified by incidents such as Stuxnet and Colonial Pipeline, which exemplify how critical infrastructure is increasingly subject to cyber threats.

CPS operators also have to meet the requirements of privacy regulations, cybersecurity standards, and governance requirements. Existing research mainly focuses on technical controls, leaving minimal guidance for integrating security with compliance. This includes frameworks like GDPR, DPDP Act 2023, NIST CSF2.0, ISO/IEC 27001 and IEC 62443 that provide obligations for privacy protection, accountability, risk management and resilience. This research seeks to fill this gap by creating the Secure and Compliant CPS Framework (SCCF).

[Figure 1](#) illustrates the evolution of Cyber-Physical Systems from traditional standalone computing environments to AI-enabled autonomous ecosystems,

highlighting the increasing integration of intelligence, connectivity, and automation.

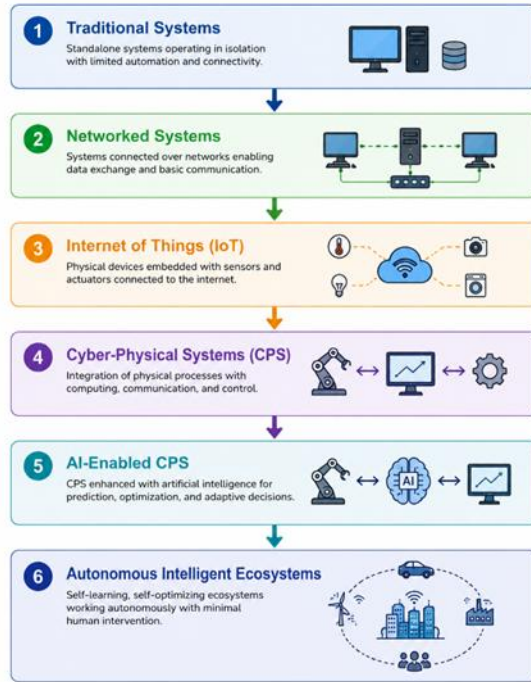


Fig. 1 Evolution of Cyber-Physical Systems

1.1 Key Contributions

The key contributions to the paper are as follows.

- Integration of cybersecurity and legal governance
- Introduction of SCCF
- Comparative legal analysis
- AI governance incorporation
- Practical roadmap for CPS organizations

2 Literature Review

Current studies about Cyber-Physical Systems (CPS) primarily discuss security risks, security of industrial control systems, and resiliency in operations. Researchers have defined major attacks on CPS environments as malware, ransomware, insider attacks, denial of service attacks (DoS), sensor spoofing,



and supply chain attacks. The rapid proliferation of devices connected to the internet has greatly enlarged the number of potential attacks against organisations, adding to their exposure to cyber risks.

There have been increased concerns regarding the protection of privacy and data governance for CPS due to the ongoing collection and processing of large amounts of private and operational data. Regulatory guidance such as GDPR and the DPDP Act stress the importance of the following concepts: transparency, accountability, lawful processing, and Privacy by Design. Recent literature on governance of AI highlights several issues surrounding the use of AI, including transparency, explainability, fairness, accountability, and the need for human supervision [4].

While frameworks including NIST CSF 2.0 [2], ISO/IEC 27001, and IEC 62443 provide valuable guidance on security, they suffer from a lack of integration when considering privacy, governance over AI, and compliance with the law. Therefore, there is still a need for a governance framework that is multidisciplinary, capable of integrating technical, legal, and organisational elements. Table 1 gives a comparative review of existing approaches.

Table 1 Comparative review of existing approaches

Author/ Framework	Security	Privacy	AI Governance	Compliance
Lee (2008) [3]	High	Low	No	No
Humayed et al. (2017) [4]	High	Low	No	No
GDPR	Medium	Very High	Partial	High
NIST CSF	High	Low	No	Medium
SCCF	High	High	High	High

3 Methodology

The research adopts a qualitative approach using three complementary methods: Systematic Literature Review (SLR), Comparative Legal Analysis (CLA), and Case Study Analysis (CSA).

In the first phase of the literature review, peer-reviewed publications were sourced from Scopus, Web of Science, IEEE Xplore, SpringerLink, and ScienceDirect. The CLA covered key regulatory and governance frameworks, including the Information Technology Act of 2000, the Digital Personal Data Protection Act 2023; General Data Protection Regulation; NIST Cybersecurity Framework version 2.0; ISO/IEC 27001; and IEC 62443. CSA analysed four representative CPS incidents, namely Stuxnet, Colonial Pipeline, Jeep Cherokee, and Medical Internet of Things (MIoT) vulnerabilities. The findings from these case studies informed the development of the proposed SCCF architecture. The overall research workflow adopted in this study is illustrated in [Figure 2](#).

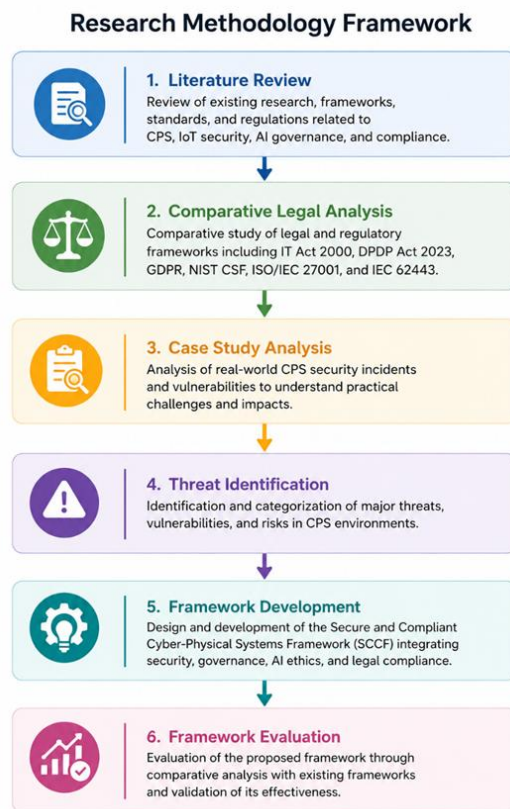


Fig. 2 Research methodology adopted for SCCF development



4 Legal Analysis

The governance of Cyber-Physical Systems is becoming increasingly affected by legal and regulatory requirements relating to protecting individuals' personal information, critical infrastructures, and public safety.

4.1 Information Technology Act

The Information Technology Act 2000 is India's primary cyber law designed to address aspects including cybercrime, unauthorised access, electronic records, and digital evidence [\[5\]](#).

4.2 Digital Personal Data Protection Act

Additionally, the Digital Personal Data Protection Act 2023 creates additional obligations regarding consent management, accountability, breach notification and on the responsibilities of data fiduciaries [\[6\]](#).

4.3 GDPR

The General Data Protection Regulations (GDPR) continue to serve as the global standard for governing privacy and creates requirements for transparency, accountability, lawfulness, purpose limitation and Privacy-by-Design [\[7\]](#).

4.4 NIST CSF

Cybersecurity standards also play an important role. The NIST Cybersecurity Framework 2.0 has established criteria that focus on governance, risk management, incident response, and resilience.

4.5 ISO/IEC 27001

ISO/IEC 27001 contains requirements that promote the establishment of an Information Security Management System (ISMS) that is structured and organised [\[8\]](#).

4.6 IEC 62443

IEC 62443 provides specific security guidance for industrial automation and operational technology environments [\[9\]](#).

[Table 2](#) compares the major legal and cybersecurity frameworks applicable to Cyber-Physical Systems. The analysis demonstrates that existing frameworks remain fragmented and fail to provide comprehensive governance guidance for

CPS ecosystems. [Figure 3](#) illustrates the primary attack vectors across different layers of Cyber-Physical Systems.

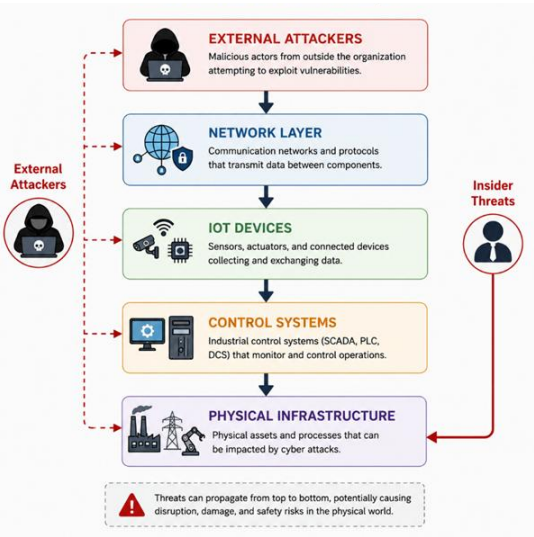


Fig. 3 Threat vectors targeting CPS ecosystems

In the last few decades, the jurisprudential and economic nature of information has been transformed, moving from its status as a tool to obtain or control external resources to one of the main resources in the current global economy. The current digital age is a constant systemic issue with protecting the privacy of an individual. The drive for personal documentation and private areas has had a precedent. Still, the interplay of globalisation and speed of technological advancement has exponentially amplified the speed and volume of personal data sharing. As a result, there has been a significant increase in regulatory clashes in the data protection landscape between public authorities and the systematised collection and treatment of personal telemetry.

Table 2 Comparative Legal Framework Analysis

Criteria	IT Act	DPDP Act	GDPR	NIST CSF	ISO 27001	IEC 62443
Privacy Protection	Low	High	Very High	Low	Medium	Low



Criteria	IT Act	DPDP Act	GDPR	NIST CSF	ISO 27001	IEC 62443
Accountability	Medium	High	High	High	High	Medium
Incident Response	Medium	Medium	High	High	High	High
Industrial Security	Low	Low	Low	Medium	Medium	Very High
AI Governance	No	Limited	Emerging	Limited	No	No

This tension came to a head in the context of Indian constitutional jurisprudence in the Supreme Court's landmark decision in the case of Justice K.S. Puttaswamy (Retd.) v. Union of India. The judiciary's decision to define the right to privacy as an inherent fundamental right secured by Article 21 of the Indian Constitution put privacy at the very heart of today's constitutional obligations. The state has a primary duty to ensure that the civil liberties and the right to digital freedom of its citizens are not subjected to arbitrary or disproportionate surveillance or exploitation of their data under this constitutional provision.

Even though it is a prominent issue in the constitution, data protection is one of the most abstract and elusive concepts in jurisprudence, and cannot be reduced to a single definition. In legal discourse, the term "data protection" is generally used to refer to the overall area of data governance, from data gathering, processing, storage to erasure of personal data. With widespread digitisation of day-to-day life, data is omnipresent and has generated great benefits for society and technology. But this advancement has engendered a fundamental split between information that people knowingly and voluntarily provide and the quietly gathered data, which is a product of ubiquitous software architectures. To counteract this imbalance, contemporary systems and frameworks strive to achieve fairness, transparency, and a rebalancing of power, putting individuals in control of how third parties may access and monetise their digital footprints (Jiang et al., 2024) [10].

The concept of "informational self-determination" is at the heart of the constitutional-privacy vs. statutory-data-protection relationship. This attitude stems from democratic ideals and gives people the right to control, at their own discretion, how, when, and what of themselves they share with others. Most

importantly, modern legal scholarship recognises that one might never have complete control over data in complex and interwoven digital infrastructures. So instead of trying to achieve complete data isolation, effective and modern regulatory frameworks seek to create a balanced, highly regulated oversight system that will achieve a balance between individual privacy rights and legitimate public administration and system needs.

Architectural evolution of Cyber-Physical Systems (CPS) has completely revolutionised the way computational processes interact with the physical world by bringing them together in an integrated network of sensors and actuators. These technologies have evolved from stand-alone networks to hyper-connected, network-dependent systems, and the need to operate within them has grown to include real-time interoperability with external networks. But all these integrated applications and services have multiplied the attack surface of the system at an exponential rate, creating important vulnerabilities in the computational and physical layers. Today, engineering techniques focus on the protection of the core hardware components, namely sensors, actuators and communications links, to reduce such risks and boost the resilience of the system. This is often done by performing real-time state estimations which emulate normal operational baselines from a single controller. This common structure allows for continual error recovery and rollback capabilities, and eventually the development of sophisticated CPS checkpointing and recovery features, which maintain physical functionality even in the presence of active compromise of the components.

Concurrently, with these engineering developments, the new level of sophistication of both the hardware and software has led to ubiquitous, multi-domain tracking across both physical and digital domains. Covert instruments like tracking cookies and hyperlinked tracking pixels can track user preferences and trends in the digital realm without the user's explicit consent. In parallel, within the physical world, the commonality of consumer internet-of-things (IoT) devices such as smartwatches, health trackers and mobile phones enables the ongoing collection of fine-grained telemetry, location and habits, which allow for the creation of more comprehensive profiles of human activity.

This surveillance-by-the-willy-nilly mode of data collection creates serious privacy and constitutional issues. The exact mechanisms that are designed to make connectivity and the increase of public convenience go hand in hand are also vectors of illegal surveillance and of systematic privacy violations. Consumer services – which are basically services that people are expecting to receive from the operation – can be easily harvested by operational telemetry for unauthorised monitoring operations. For example, if the cameras are smart residential entryways, which are intended to be used to verify identity for visitors at a local level, there is a risk of exploitation through an unauthorised third party gaining access, which could lead to the exposure of localised identity and temporal-spatial information. These household data streams, when combined across large time periods and/or across many different distributed



nodes, enable mass public surveillance. These intrusive techniques often run in parallel to the rights of consumers to give positive consent, and present substantial regulatory risk for data exposure in transit and storage.

As a result, the ability to have autonomous control over personal information has now become a pressing need in the current CPS deployments from a jurisprudential and technical standpoint. As smart devices proliferate and increasingly become part of everyday life and public spaces, there must be strong legal and structural mechanisms that enable users to have meaningful control over the life cycle of their data – its storage, dissemination, and permanent destruction. To reach this level of digital self-determination, it is important that the interfaces used are transparent and that consent is given in a clear and unambiguous manner, so that the person can make an informed choice. Safeguarding data integrity during data movement and rest is also a major aspect of data privacy – and is no simple task given the widely varying types of processing and cryptographic power of different IoT device hardware.

5 Threat Analysis

The threat landscape for contemporary Cyber-Physical Systems (CPS) is highly intricate. Some of the highest risk factors involve: malware; malicious software; malware; viruses; cybercrime; cyberattacks; cyber espionage; cyber sabotage; and Artificial Intelligence (AI) manipulated attacks. [Table 3](#) summarises the qualitative threat severity assessment.

Table 3 CPS Threat Severity Matrix

Threat	Likelihood	Impact	Risk Level
Malware	High	High	High
Ransomware	High	Critical	Very High
Sensor Spoofing	Medium	Critical	High
Insider Threat	Medium	High	High
Supply Chain Attack	Medium	High	High
AI Manipulation	Medium	High	High

The qualitative ratings (Low, Medium, High, and Very High) presented in the Threat Severity Matrix were assigned through expert judgment based on the estimated likelihood of occurrence and the potential impact of each threat on confidentiality, integrity, availability, operational continuity, public safety, and regulatory compliance. These ratings are intended as conceptual qualitative assessments derived from the literature review, comparative legal analysis, and representative CPS case studies, rather than statistically derived quantitative measurements.

The Stuxnet malware attack is a well-known example that demonstrated how malware can exploit and manipulate industrial control systems to cause physical damage. The recent Colonial Pipeline ransomware attack is another example of the growing threats to critical infrastructure by cybercriminals. Another example of the types of risks associated with cyber-physical systems is illustrated by the Jeep Cherokee hack; a demonstration of the dangers associated with automobile/cyber-physical systems.

In summary, when security measures fail within a cyber-physical system, multiple issues are created, including: legal liability; monetary losses; regulatory scrutiny; and issues related to public safety. [Table 4](#) presents the qualitative risk heat matrix used for prioritising CPS threats.

Table 4 CPS Risk Heat Matrix

Impact ↓ / Likelihood →	Low	Medium	High
Critical	-	Sensor Spoofing	Ransomware
High	-	Insider Threat AI Manipulation Supply Chain	Malware
Medium	-	-	-
Low	-	-	-

6 Secure and Compliant Cyber-Physical Systems Framework (SCCF)

The SCCF is a multilayered governance system that links Cyber Security, Privacy, Governance of Artificial Intelligence, and Legal Regulation.

The framework consists of five layers:

- (1) Physical Security Layer
- (2) Cyber Security Layer
- (3) Data Governance Layer
- (4) AI Governance and Ethical Layer
- (5) Legal and Regulatory Compliance Layer

As a result, the framework is built to facilitate the implementation of Security by Design, Privacy by Design, Compliance by Design, and Governance by Design principles. [Figure 4](#) presents the proposed Secure and Compliant Cyber-Physical Systems Framework (SCCF). [Figure 5](#) demonstrates how legal obligations are translated into governance policies, security controls, compliance evidence, and audit verification. [Figure 6](#) summarises the fundamental principles of AI governance adopted within the proposed SCCF.

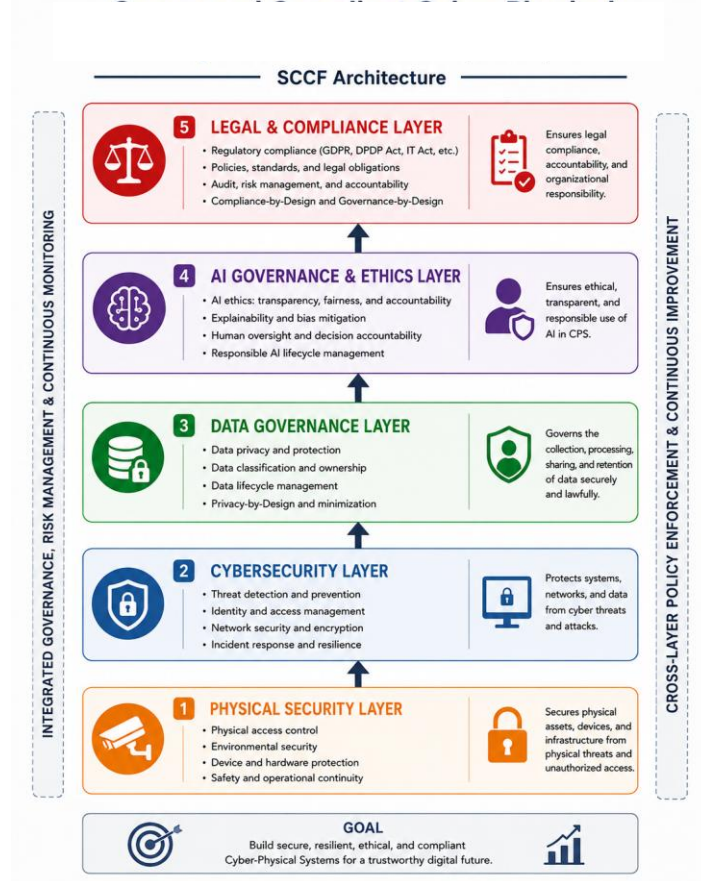


Fig. 4 Secure and Compliant Cyber-Physical Systems Framework (SCCF)



Fig. 5 SCCF Compliance Mapping Model

[Table 5](#) depicts a comparison of SCCF with existing frameworks. The five governance layers operate in a hierarchical yet interdependent manner. Physical Security provides the foundation upon which cybersecurity controls protect digital assets. Data Governance regulates information throughout its lifecycle, while AI Governance ensures ethical, transparent, and accountable decision-making. The Legal and Compliance Layer continuously oversees all operational layers by monitoring regulatory obligations, conducting audits, and ensuring organizational accountability. In practice, trade-offs such as transparency versus operational security are addressed through risk-based governance policies and proportional compliance mechanisms.



Fig. 6 AI Governance and Ethics Model

Table 5 SCCF vs Existing Frameworks

Capability	NIST CSF	ISO 27001	IEC 62443	SCCF
Cybersecurity	✓	✓	✓	✓
Privacy Protection	Partial	Partial	No	✓
AI Governance	No	No	No	✓

Capability	NIST CSF	ISO 27001	IEC 62443	SCCF
Legal Compliance	Partial	Partial	Partial	✓
Unified Governance	No	No	No	✓

7 Illustrative Application of SCCF

Consider a smart hospital deploying interconnected medical devices, AI-assisted diagnostics, and cloud-based patient monitoring. Under the SCCF, physical security safeguards protect medical equipment from unauthorised access, cybersecurity controls secure communications and connected devices, data governance ensures lawful processing of patient information under the DPDP Act and GDPR, AI governance promotes explainability and human oversight of diagnostic recommendations, while the legal and compliance layer continuously monitors regulatory obligations, audit trails, and breach notification requirements. This example demonstrates how SCCF integrates technical, organisational, and legal controls into a unified governance framework for critical healthcare infrastructure.

8 Evaluation and Discussion

SCCF was compared to NIST CSF 2.0, ISO/IEC 27001 and IEC 62443 using a comparative analysis for evaluation. The results of this evaluation showed that the SCCF provides broader governance coverage due to its integration of cyber security, privacy protection, artificial intelligence governance and legal compliance into one overall architecture.

The SCCF Framework has contributed to both theory and practice. It has contributed to theory through the extension of the literature on governance of Cyber-Physical Systems (CPS), and it has provided organisations with a roadmap to implement secure and compliant CPS ecosystems. The SCCF Framework will also provide accountability, transparency and readiness for regulations.

The implementation of SCCF may differ across organizations depending on their size, sector, regulatory obligations, technological maturity, and available resources. Small and medium-sized enterprises may adopt the framework incrementally, whereas critical infrastructure operators may implement all governance layers as part of comprehensive enterprise risk management programs.



[Table 6](#) compares the proposed SCCF with existing governance frameworks. This Table demonstrates that the proposed SCCF provides broader governance coverage than existing cybersecurity frameworks by integrating cybersecurity, privacy protection, AI governance, and legal compliance into a unified governance architecture. Unlike traditional frameworks, SCCF adopts a multidisciplinary approach that simultaneously addresses technical, organizational, ethical, and regulatory requirements for Cyber-Physical Systems.

Table 6 Comparative Evaluation of SCCF and Existing Frameworks

Framework	Cyber	Privacy	AI	Legal
NIST	✓	Partial	✗	Partial
ISO	✓	Partial	✗	Partial
IEC	✓	✗	✗	Partial
SCCF	✓	✓	✓	✓

9 Limitations and Future Work

The proposed SCCF is primarily a conceptual governance framework derived through qualitative analysis of literature, legal instruments, and representative case studies. Although the framework provides a comprehensive governance perspective, it has not yet been empirically validated through industrial deployment, expert evaluation, or quantitative performance assessment. Future work should focus on validating the framework in real-world CPS environments, developing governance maturity models, and integrating automated compliance monitoring mechanisms.

10 Conclusion

Cyber-Physical Systems (CPSs) are an important part of the modern digital infrastructure. With the increase in convergence of Artificial Intelligence (AI) and the Internet of Things (IoT) with other technologies, the growing complexity of these systems has resulted in new challenges associated with security, privacy, governance, and compliance. Presently existing frameworks are a good source of information for organisations but are not sufficiently integrated regarding their treatment of cybersecurity and legal obligations. The purpose of this research is to introduce the Secure and Compliant Cyber-Physical Systems Framework (SCCF), which consists of five layers of governance focused on Physical Security, Cybersecurity, Data Governance, AI Governance, as well as Legal & Regulatory Compliance. The SCCF provides organisations with the ability to better align their technical controls to the

requirements of the law and provides a foundation for Building Resilient and Trustworthy Cyber-Physical Systems (CPS). Overall, the proposed SCCF provides a comprehensive governance roadmap capable of supporting the development of secure, trustworthy, resilient, and legally compliant Cyber-Physical Systems for future intelligent digital infrastructures.

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Contribution of each Author

Sunetra Chatterjee: Conceptualisation, Research design, SCCF framework development, Cybersecurity analysis, Methodology, Manuscript review and editing.

Akash Chatterjee: Legal analysis, Literature review, Comparative legal framework analysis, Case law interpretation, Draft preparation, Reference compilation.

Conflict of Interest Statement

The authors declare no conflict of interest.

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